Object Oriented Programming (CST-004) - Unit 4

Unit 4 - Templates & Exception Handling - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: templates for generic code and exception handling for runtime errors.

Key ideas: A **function template** is one function that works for any type: template <class T> T add(T a, T b). A **class template** generalises a whole class (like a Stack<T> for int or string). The compiler generates a concrete version for each type you use - this is compile-time (static) polymorphism. **Exception handling:** code that may fail goes in **try**, an error is signalled with **throw**, and a matching **catch** block handles it. catch(...) catches anything. If no handler matches, the program terminates. You can rethrow from a catch and write your own exception classes.

Exam pointers: expect one program using a function template (max/swap/sort for two types) and one using try/throw/catch - divide-by-zero is the usual example. Theory: "templates vs macros" and "why exceptions beat error codes" are common short answers.

Made by Anas. Free forever. Watch the linked videos on the site after reading.